GyroMem: Curvature-Adaptive Memory for Long-Horizon LLM Agents

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Abstract

Long-horizon LLM agents need memory that persists across sessions, organizes itself under use, forgets gracefully, and serves retrieval at low cost. The last year of research has converged on multi-relational memory graphs, retrieval-driven consolidation, and the separation of memory control from generation. Yet every system in the recent agentic-memory wave stores memories as Euclidean vectors, compares them with cosine similarity, and indexes them with Euclidean approximate nearest-neighbor structures. Curved geometry has touched retrieval-augmented generation [20] and unreviewed engineering projects, but not—to our knowledge—the full lifecycle of agent memory with formal mechanisms. This paper proposes **GyroMem**, a theory of agentic memory in which memories live on a *curved* manifold: a product of hyperbolic, spherical, and Euclidean components whose curvatures are learned per memory region. Retrieval becomes geodesic distance; consolidation becomes geodesic transport of memory points toward useful queries, with confidence-bounded step sizes that yield stability guarantees absent from current heuristic and LLM-driven update rules; forgetting becomes reversible geometric drift rather than deletion; and the transient (KV-cache) and persistent (external store) tiers are managed under a single geometric budget. We formalize the theory, state the exact gyrovector operations it requires, prove a stability sketch for the reconsolidation update, and pre-register a falsification-oriented experimental program—including explicit kill criteria—under which the theory should be abandoned or revised. This is a theory paper: we claim no empirical results for GyroMem itself. Our contribution is a precise, implementable, and falsifiable geometric reframing of agentic memory, plus an honest accounting of where it is most likely to fail.

Keywords: agentic memory, hyperbolic geometry, product manifolds, retrieval-augmented generation, memory consolidation, geodesic reconsolidation

1 Introduction

An LLM agent that cannot remember is an amnesiac savant: brilliant within a context window, a stranger to everything before it. As agents take on long-horizon work—software projects spanning weeks, personal assistants spanning years, scientific collaborators spanning careers—memory stops being a feature and becomes the substrate. The research community has responded with remarkable energy: in a single recent month (September–October 2026), at least twelve papers advanced agentic memory, from control-plane architectures to KV-cache eviction to consolidation governance [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12].

Stepping back from this burst of activity, a pattern—and a conspicuous absence—emerges. The pattern: memory is becoming a *closed-loop lifecycle* rather than a store. Retrieval rewires the graph [3, 7]; context is constructed just-in-time at query time rather than pre-compressed [2]; memory control is being split off from generation into cheaper dedicated machinery [1];

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forgetting is becoming a first-class, principled design variable [7, 5]. The absence: *the geometry never changed.* Every system we surveyed stores memories as Euclidean vectors, scores them with cosine similarity (or fusions thereof), and indexes them with Euclidean ANN structures (HNSW, PQ). The vector mathematics of memory—the space itself in which memories live, move, and decay—has been treated as fixed plumbing while all the innovation happened one level up.

This paper asks whether the plumbing is the problem. Agent memory is simultaneously *hierarchical* (identity → preferences → episodes → raw facts), *relational* (entities linked in cycles and mutual reference), and *flat* (surface feature similarity). Euclidean space is a poor host for the first of these: trees embed into Euclidean space only with severe distortion, while hyperbolic space accommodates them with arbitrarily low distortion [13]. The mismatch is not aesthetic. If memories live in the wrong geometry, then every downstream operation—retrieval ranking, consolidation updates, forgetting schedules—must compensate with heuristics, thresholds, and expensive LLM-driven repairs for what the geometry gets wrong for free.

Thesis. *Agentic memory should live on a curved manifold: a product of hyperbolic, spherical, and Euclidean components with learned per-region curvature. Retrieval is geodesic distance. Consolidation is geodesic transport. Forgetting is geometric drift. The resulting system is cheaper than LLM-driven memory surgery, more stable than heuristic decay, and falsifiable in a single round of ablations.*

We name this theory **GyroMem**, after the gyrovector spaces that furnish its algebra. Our contributions are:

1. A formal specification of a product-manifold memory space for LLM agents, with explicit formation (exponential map), retrieval (geodesic distance), consolidation (confidence-bounded geodesic transport), promotion (radial motion), and forgetting (reversible drift) operators.

2. A stability sketch for the reconsolidation update, giving the kind of convergence argument that usage-driven plasticity in current systems lacks.

3. A unification of the transient (KV-cache) and persistent (external) memory tiers under one geometric budget, addressing a split no current system bridges.

4. A pre-registered, falsification-oriented experimental program with named hypotheses, ablations that isolate each geometric claim, and explicit kill criteria—so the theory can fail fast and honestly.

5. An candid failure analysis: where the theory is most likely wrong, what prior art constrains it, and what a negative result would still teach.

We emphasize what this paper is *not*: it reports no experiments on GyroMem, claims no benchmark victories, and invents no data. It is a theory paper written *before* the experiments, in the open, so that the subsequent empirical program—pass or fail—can be judged against pre-committed predictions rather than retrofitted narratives.

2 Background and Related Work

2.1 Agentic memory systems: the state of play

The recent literature clusters into several lines. **Control-plane architectures** separate memory decisions from generation: Jev-Mem routes all high-frequency memory decisions (typing, routing, budget allocation, traversal, stopping) through a lightweight typed-probabilistic controller, reserving the LLM for synthesis, reaching 0.777 on LoCoMo (+11.0% relative over MAGMA) with 6.6× faster builds and −36.7% latency [1]. **Just-in-time memory** rejects ahead-of-time

compression: JAM preserves complete raw histories in a hierarchical page-store and performs runtime agentic research per query, with a 4B model outperforming 7–14B trained memory agents [2]. **Retrieval-driven reconsolidation** treats recall as a write event: REALM applies LLM-driven edge surgery (add/strengthen/weaken with confidence-bounded updates) on a heterogeneous cognitive graph after retrieval [3]; EngramRAG implements a waking/dreaming cycle with Hebbian edge plasticity, usage-modulated PageRank, and topology-aware decay [7]. **KV-cache management** optimizes the transient tier: ActKV exploits stable action-critical attention patterns [4], while AgentKV models the phase-mixture of query subspaces [10]. **Index elasticity** arrives via Matryoshka-nested codes serving multiple operating points from one artifact [6, 9]. **Governance and evaluation** mature: the Epistemics program exposes degenerate forgetting, proves that consolidation-quality metrics do not predict real transfer ($\rho = -0.24$), and proposes governed consolidation with poison-resistance, reversibility, and auditability [5]; Bio-Memory conditions retrieval on biometric identity [8]. Differentiable memory takes another route: δ-mem maintains a compact 8×8 associative state updated by delta-rule learning, read out as low-rank corrections to a frozen backbone's attention [24].

Common to all twelve: Euclidean embeddings, cosine similarity, Euclidean ANN. Within this wave, the geometry is never varied, never ablated, never questioned.

2.2 Hyperbolic representation learning

The case for non-Euclidean representation is well established outside agentic memory. Nickel and Kiela showed that hierarchical data—taxonomies, trees, entailment cones—embeds into the Poincaré ball with far lower distortion than into Euclidean space of the same dimension, because volume in hyperbolic space grows exponentially with radius, matching the exponential growth of tree nodes with depth [13]. Ganea, Bécigneul, and Hofmann extended the toolkit to hyperbolic neural networks, defining Möbius versions of addition, matrix multiplication, and recurrent updates, with closed-form exponential and logarithmic maps [14]. Gu et al. generalized further to *product manifolds*—Cartesian products of hyperbolic, spherical, and Euclidean components with learned curvatures—showing that heterogeneous data is best served by heterogeneous geometry [15]. These results are our direct mathematical foundation; we apply them to the full lifecycle of *agent* memory rather than to static embedding tasks. The nearest existing efforts are discussed in §2.4.

2.3 Nearest-neighbor search beyond Euclidean space

If memories live on a manifold, retrieval needs manifold-aware search. Hyperbolic nearest-neighbor methods exist but are immature relative to FAISS/HNSW. Graph-based search has been shown to enjoy *lower* time complexity in hyperbolic than in Euclidean space in the dense regime [22]; the first LSH construction native to hyperbolic space appeared recently [23]; and the working engineering pattern is two-stage Euclidean pre-filtering with hyperbolic reranking [19]. Tangent-space approximations that map neighborhoods into the tangent plane at the query point complete the toolkit. We treat ANN on the product manifold as an engineering problem with known partial solutions, not as a solved one—a risk we carry explicitly into §6.

2.4 Curved geometry for retrieval and memory: the nearest neighbors

Cao et al. introduce HyperbolicRAG, embedding graph-RAG nodes in a shared Poincaré manifold with mutual-ranking fusion of Euclidean and hyperbolic signals, improving multi-hop QA over Euclidean baselines [20]—but it is retrieval-only: no memory lifecycle, no reconsolidation, no forgetting, single curvature. Chakrabarti projects pretrained embeddings into the Lorentz model for edge-device RAG reranking [21]. The closest effort to GyroMem is GAMI, an engineering

project implementing long-term agent memory on a product manifold $\mathbb{H}^{32} \times \mathbb{S}^{16} \times \mathbb{R}^{64}$ with two-stage retrieval and a dream-cycle consolidation loop [19]; its reported gains are unverified, and it proposes no geodesic reconsolidation rule, no stability analysis, and no forgetting dynamics. To our knowledge, no peer-reviewed work combines product-manifold geometry with retrieval-driven reconsolidation, reversible forgetting, and unified tier budgeting for agent memory. GyroMem's contribution is therefore (i) the first formal treatment of curved-geometry agent memory, with (ii) three mechanisms absent from all prior art we found: geodesic reconsolidation with a contraction-stability argument, curvature-parameterized reversible forgetting, and tier-unified budgeting—plus (iii) a pre-registered falsification program.

2.5 Neuroscience: complementary systems and reconsolidation

Two ideas from neuroscience recur, usually as loose metaphor, across the agent-memory literature; we attempt to make them literal. Complementary Learning Systems theory holds that the hippocampus (fast, episodic) and neocortex (slow, semantic) consolidate via replay [17]—the template for waking/dreaming splits [7]. Memory reconsolidation holds that recalled memories enter a labile state in which they can be strengthened, weakened, or rewritten [18]—the template for REALM's post-retrieval updates [3]. GyroMem's geodesic transport is reconsolidation made geometric: the labile state is motion along the manifold, and its bounds are step sizes, not prompts.

2.6 The evaluation problem

A theory of memory is only as good as its measures, and the measures are troubled. LoCoMo with LLM-as-a-judge dominates evaluation, yet the Epistemics program demonstrates that well-constructed consolidation-quality scores do not predict real transfer accuracy [5], and retrieval-metric wins need not transfer to downstream generation [7]. Any honest experimental program must therefore measure *transfer*—whether consolidated memory changes what the agent can do—not just recall. We build this lesson into our pre-registration (§5).

3 Motivation: four gaps

From the survey above we distill four gaps that GyroMem is designed to close. Each is a falsifiable claim about the literature, and each maps to one component of the theory.

Gap 1: the geometry is fixed and unexamined. No system in the surveyed wave varies the embedding geometry; cosine similarity on Euclidean vectors is universal there. Curved geometry exists at the edges—retrieval-only systems [20, 21] and unreviewed engineering [19]—but not as a formal, evaluated theory of the memory lifecycle. Yet agent memory is conspicuously hierarchical, the exact regime where hyperbolic geometry dominates [13]. *Component:* product-manifold memory space (§4.1).

Gap 2: plasticity without theory. Usage-driven updates—Hebbian rules, topology-aware decay, LLM graph surgery—are hand-tuned or prompt-driven, with no convergence or stability analysis. Contrast the rigorous capacity theory of classical associative memory. *Component:* geodesic reconsolidation with a contraction argument (§4.4).

Gap 3: two tiers, no joint management. KV-cache systems [4, 10] and external-memory systems [1, 3, 2] optimize disjoint budgets though they compete for the same serving resources. *Component:* unified geometric budget (§4.7).

Gap 4: forgetting as an afterthought. Decay is wall-clock exponential (refuted as a retention rule [7]), thresholded hand-tuning, or—worst—a budget release valve that destroys load-bearing memories [5]. *Component:* forgetting as reversible geometric drift (§4.5).

4 The GyroMem Theory

4.1 The product-manifold memory space

Definition 1 (Memory manifold)**.** *Let*

$$\mathcal{M} = \mathbb{H}^a_{-c_h} \times \mathbb{S}^b_{c_s} \times \mathbb{R}^c, \tag{1}$$

the Cartesian product of a hyperbolic component of dimension a and curvature $-c_h < 0$, a spherical component of dimension b and curvature $c_s > 0$, and a Euclidean component of dimension c. Each memory m is a point $m = (m_h, m_s, m_e) \in \mathcal{M}$. Curvatures c_h, c_s are learned per memory region *(see below), not fixed globally, following [15].*

The three components have distinct duties. The **hyperbolic component** hosts hierarchical content: identity facts near the origin, preferences further out, episodes near the boundary—so that abstraction level is literally radial position. The **spherical component** hosts relational content: entity–entity associations and temporal cycles, where closure (returning to the start) is natural. The **Euclidean component** hosts flat featural content where cosine-style similarity is already adequate. Distance on $\mathcal{M}$ is the weighted product distance

$$d_{\mathcal{M}}(p,q)^2 = \alpha_h\, d_{-c_h}(p_h, q_h)^2 + \alpha_s\, d_{c_s}(p_s, q_s)^2 + \alpha_e\, \|p_e - q_e\|^2, \tag{2}$$

with learnable non-negative weights $\alpha_h, \alpha_s, \alpha_e$. A *region* is a subset of memories sharing curvature parameters and weights (e.g., "user preferences," "project episodes"); regions are formed by clustering in the tangent space at the origin and may split or merge under the dreaming consolidation described in §4.5.

Remark 1 (Why products, not pure hyperbolic)**.** *Pure hyperbolic memory would repeat the monoculture it criticizes. Relational cycles embed poorly in hyperbolic space and naturally in spherical space; flat features need no curvature at all. The product follows [15]: let the data's heterogeneity choose the geometry. If ablations show one component dominates, the theory simplifies—that is a feature, not a retreat.*

4.2 Memory formation: the exponential map

New memories arrive as Euclidean vectors (encoder outputs). Formation maps them onto $\mathcal{M}$ via the exponential map at the origin of each component. For the Poincaré ball $\mathbb{B}^n_c$ of curvature $-c$, with Möbius addition $\oplus_c$ (formulary in Appendix A),

$$\exp_0^c(v) = \tanh(\sqrt{c}\,\|v\|)\,\frac{v}{\sqrt{c}\,\|v\|}, \qquad \log_0^c(y) = \operatorname{artanh}(\sqrt{c}\,\|y\|)\,\frac{y}{\sqrt{c}\,\|y\|}. \tag{3}$$

The norm of the pre-image v controls radial placement: small $\|v\|$ lands near the origin (abstract), large $\|v\|$ near the boundary (specific). Formation therefore requires a *calibrated* encoder norm, which we obtain by training the encoder with a radial regularizer that aligns $\|v\|$ with an abstraction label (identity < preference < episodic < verbatim). This is the one trained component of the otherwise training-free theory, and its necessity is itself a falsifiable claim (H2 in §5).

4.3 Retrieval as geodesic distance

Definition 2 (GyroMem retrieval)**.** *Given query $q \in \mathcal{M}$ (formed as in §4.2), return the k memories minimizing $d_{\mathcal{M}}(q, m)$, optionally fused across regions by reciprocal-rank fusion over per-region rankings.*

Hyperbolic distance expands exponentially near the boundary of the ball: $d_c(u, v)$ grows as $\approx \frac{2}{\sqrt{c}} \log \frac{1}{1-r}$ for points at radius r. A query formed at the abstraction level of "preferences" is therefore geodesically near the preference region and exponentially far from individual episodes—*hierarchical pre-filtering falls out of the metric*, where Euclidean systems must learn or prompt it. Spherical distance handles cyclic association ("A relates to B relates to C relates to A") without the hub distortion flat spaces induce.

4.4 Reconsolidation as geodesic transport

This is the load-bearing mechanism. Where REALM performs LLM-driven graph surgery after retrieval [3], GyroMem moves points.

Definition 3 (Geodesic reconsolidation)**.** *Let $m \in \mathcal{M}$ be a retrieved memory, $q \in \mathcal{M}$ the query (or the retrieved context centroid), and $c \in [0, 1]$ a confidence score for the retrieval's usefulness (e.g., downstream answer entailment, or a learned critic). The update is*

$$m \leftarrow \exp_m(\eta\, c \cdot \log_m(q)) \qquad \textit{(strengthen)}, \tag{4}$$

$$m \leftarrow \exp_m(-\eta\, c \cdot \log_m(q')) \qquad \textit{(weaken, against contradicted content } q'\textit{)}, \tag{5}$$

with step size $\eta \in (0, 1)$. In the Poincaré component this is Möbius scalar multiplication along the geodesic; in the Euclidean component it reduces to the familiar $m \leftarrow (1 - \eta c)m + \eta c\, q$.

The update is $O(d)$ per memory—one logarithmic and one exponential map—versus an LLM call per edge in surgery-based approaches. It is also the precise geometric analogue of REALM's $w \leftarrow \mathrm{Clip}(w + \eta c)$: bounded, confidence-gated, topology-preserving.

Proposition 1 (Stability sketch)**.** *Fix a target q and consider the strengthen map $T(m) = \exp_m(t \cdot \log_m(q))$ with $t = \eta c \in [0, 1)$ on a Hadamard manifold (simply connected, non-positive curvature; the hyperbolic component qualifies). Then T is a strict contraction toward q: $d(T(m), q) = (1 - t)\, d(m, q)$. Iterating T converges to q geometrically; with confidence-weighted, query-varying targets the process is a stochastic approximation whose step sizes ηc_t satisfy the usual Robbins–Monro conditions when $\sum_t \eta c_t = \infty$ and $\sum_t (\eta c_t)^2 < \infty$, yielding almost-sure convergence to the confidence-weighted Fréchet mean of the query distribution.*

Sketch. The contraction identity is the defining property of geodesic interpolation on any uniquely geodesic space [14]. The stochastic-approximation claim follows from standard results for incremental Fréchet-mean estimation on Hadamard manifolds, under bounded curvature and compact support assumptions we state as explicit preconditions. A full proof with the product-manifold (mixed-sign curvature) case, where the spherical component is *not* Hadamard, is deferred to future work— this is the honest boundary of the claim. □

Remark 2 (What the sketch does and does not buy)**.** *It buys: no runaway drift under repeated retrieval (the failure mode of unbounded Hebbian rules), and a principled step-size schedule. It does not buy: guarantees under adversarial queries (governance, per [5], is still required), or anything about the spherical component, where geodesics are not unique. We state this plainly because the literature's habit is to over-claim exactly here.*

4.5 Forgetting as reversible geometric drift

Wall-clock exponential decay $e^{-\lambda \Delta t}$ is refuted as a retention rule [7]: it evicts foundational facts. Hand-tuned thresholds are fragile [5]. GyroMem replaces both with drift on the manifold. Each memory carries a usage potential $u(m, t)$ decaying with disuse and refreshed by retrieval. The drift field is

$$\frac{dm}{dt} = -\lambda(u)\, \nabla_m\, \rho(m), \tag{6}$$

where $\rho(m)$ is a radial potential (distance from the region's abstraction centroid in the hyperbolic component) and $\lambda(u)$ is monotone decreasing in u. Low-usage memories drift *inward*: episodes collapse toward their summaries (promotion, not deletion); continued disuse drifts them past an inner horizon (true forgetting). Crucially, drift is a diffeomorphism-integrable flow: re-access applies the reverse transport, so forgetting is *reversible by construction*—the governance property G2 (reversibility) of [5] falls out of the mathematics rather than being bolted on. Nothing is ever deleted except by explicit, audited user action.

4.6 Promotion as radial motion

The consolidation ladder of [5]—keep / summarize / extract skill / distill rule / forget—becomes radial position. Dreaming-phase consolidation (offline, à la [7]) clusters memories in tangent space and moves cluster centroids inward while pushing members outward slightly (discrimination), i.e., abstraction is centripetal motion. A "skill" is a centroid that has drifted far enough inward to serve as a retrieval attractor for its former members. No separate promotion policy with tunable thresholds is needed; there is one drift field and its inverse.

4.7 Unified tier budgeting

KV-cache systems [4, 10] and external stores [1, 3, 2] manage disjoint budgets. In GyroMem, the transient tier is a high-curvature, small-radius region of the *same* manifold $\mathcal{M}$ (recent tokens live near the query point; curvature is high so distances discriminate sharply at small scales), while the persistent tier is the low-curvature bulk. A single controller allocates a global memory budget B across tiers and regions by expected utility per unit cost, generalizing Jev-Mem's u_d utility estimates [1]:

$$\max_{\{b_r\}} \sum_r U_r(b_r) \quad \text{s.t.} \quad \sum_r b_r \le B, \tag{7}$$

where U_r is the estimated marginal retrieval utility of region r at allocation b_r. Eviction in the transient tier is then not a separate heuristic but the same drift dynamics at a faster timescale.

4.8 Just-in-time geodesic research

Following JAM's refutation of ahead-of-time compression [2], GyroMem constructs query-time context by *navigating* $\mathcal{M}$: starting from $\exp_0(\text{query})$, expand a geodesic ball, assess sufficiency (has the ball's Fréchet variance fallen below τ?), and stop—a geometric stopping rule replacing hand-tuned thresholds. Raw histories are preserved (JAM's Memorizer lesson); the manifold is the index, not a compression of it.

5 Proposed experimental validation

This section pre-registers the empirical program. *No results are reported*—that is the point. Each hypothesis names its ablation, its metric, and the condition under which we abandon the corresponding claim.

Benchmarks. LoCoMo and LongMemEval (the field standards), plus a *transfer benchmark* we will construct following the Epistemics warning [5]: tasks where consolidated memory must change what the agent can *do* (skill transfer, contradiction recovery), scored against a computable oracle, not an LLM judge alone.

Hypotheses and kill criteria.

H1 (geometry matters). Manifold+geodesic retrieval beats the *identical* system with Euclidean embeddings + cosine on recall@k and on transfer, at matched dimension. *Kill:* no statistically significant win on both; pivot to Euclidean geodesic updates (still novel vs. LLM surgery on cost) or abandon.

H2 (calibrated formation). The radial regularizer (§4.2) is necessary: without it, hierarchical pre-filtering vanishes. *Kill:* ablated formation matches full formation; drop the trained component and simplify the theory.

H3 (cheap consolidation). Geodesic reconsolidation matches or exceeds LLM graph surgery (REALM-style) on consolidation quality at $<1\%$ of the update cost. *Kill:* quality gap at fixed budget; retain geometry for retrieval only.

H4 (reversible forgetting). Geometric drift shows zero catastrophic forgetting of load-bearing facts under budget pressure (where [5] documents 74% destruction by baselines), with full recovery on re-access. *Kill:* any irreversible loss; the drift field is wrong and must be redesigned.

H5 (unified budget). Joint tier budgeting beats separate KV-cache + external-store management on end-to-end task throughput at fixed total memory. *Kill:* no win; keep the tiers separate and the manifold for the external store only.

Methodological commitments. Fixed random seeds; paired bootstrap confidence intervals; ablations that change exactly one component; public release of code, configs, and the transfer benchmark; independent replication invited before any product claims. We will publish negative results with the same prominence as positive ones.

6 Discussion: where this is most likely to fail

1. The hierarchy assumption. Hyperbolic geometry wins when data is hierarchical [13]. Agent memories may be messier—flat lists of facts, adversarial noise, contradictory updates—in which case the manifold is expensive ornament. H1 is designed to detect exactly this.

2. ANN immaturity. Product-manifold nearest-neighbor search lacks FAISS-grade tooling. Our fallback stack—tangent-space candidate generation at the query point plus exact geodesic rerank—is $O(\log n)$ for candidates but exact rerank is linear in the candidate set; at billion-scale this may lose to Euclidean HNSW on latency. This is an engineering risk, not a theoretical one, and we carry it openly.

3. The spherical component. Non-unique geodesics, no contraction guarantee, curvature-sign mixing without a full theory (noted in the Proposition's proof sketch). If the spherical component causes more trouble than value, the honest move is a hyperbolic–Euclidean product, and we have pre-committed to reporting that.

4. Encoder calibration. H2's radial regularizer is the theory's one trained component and therefore its most fragile: if abstraction level cannot be calibrated into embedding norm, formation collapses to an arbitrary projection and the hierarchy story fails.

5. Evaluation gaming. Beating LoCoMo is not the goal; transfer is. If our transfer benchmark turns out to be gameable or circular, H1–H5 prove nothing. We will adversarially probe our own benchmark before trusting it, following the anti-leakage discipline of [5].

6. Stronger null hypotheses. The fairest baselines are not mem0-style systems but the best 2026 systems (Jev-Mem, JAM, REALM) with *our* evaluation harness. A win against weak baselines is not a win.

7. Prior art may narrow the contribution. GAMI [19] already engineers product-manifold agent memory; if its reported gains replicate under independent evaluation, GyroMem's novelty narrows to formalization, the three mechanisms, and the evaluation program. We would welcome that result—it would confirm the geometry bet while disciplining our claims.

7 From theory to system: engineering roadmap

Briefly, since this paper's job is the theory: (0) prior-art sweep and formal spec—this paper; (1) Python prototype (Poincaré/Lorentz store, gyrovector ops, drift dynamics) on a LoCoMo

subset; (2) full evaluation per §5 with kill-criteria gates; (3) if gates pass, open-source engine, public benchmark release, hosted API. Each phase is gated; phases do not proceed on hope.

8 Conclusion

GyroMem reframes agentic memory geometrically: a product manifold with learned curvature, geodesic retrieval, reconsolidation as geodesic transport, forgetting as reversible drift, and one budget across tiers. It is motivated by four verified gaps in the recent literature, grounded in established mathematics, and—unusually for a theory paper—shipped with its own falsification program. It may be wrong; §6 explains exactly how. If it is right, memory stops being a pile of vectors with heuristics on top and becomes what it should have been all along: a space with a shape, in which remembering is moving, learning is transport, and forgetting is drift that can be undone.

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A Gyrovector formulary

We follow the convention of [14]: the Poincaré ball $\mathbb{B}_c^n = \{x \in \mathbb{R}^n : c\|x\|^2 < 1\}$ has curvature $-c < 0$. Möbius addition is

$$x \oplus_c y = \frac{(1 + 2c\langle x, y\rangle + c\|y\|^2)\, x + (1 - c\|x\|^2)\, y}{1 + 2c\langle x, y\rangle + c^2\|x\|^2\|y\|^2}. \tag{8}$$

Möbius scalar multiplication by $r \in \mathbb{R}$:

$$r \otimes_c x = \tanh\big(r \,\operatorname{artanh}(\sqrt{c}\|x\|)\big)\, \frac{x}{\sqrt{c}\|x\|}. \tag{9}$$

The induced distance is

$$d_c(x, y) = \frac{2}{\sqrt{c}} \operatorname{artanh}\big(\sqrt{c}\,\|{-x} \oplus_c y\|\big), \tag{10}$$

equivalently $d_c(x,y) = \frac{1}{\sqrt{c}} \operatorname{arcosh}\Big(1 + \frac{2c\|x-y\|^2}{(1-c\|x\|^2)(1-c\|y\|^2)}\Big)$. The exponential and logarithmic maps at x are

$$\exp_x^c(v) = x \oplus_c \left(\tanh\left(\sqrt{c}\,\frac{\lambda_x^c\|v\|}{2}\right)\frac{v}{\sqrt{c}\,\|v\|}\right), \qquad \lambda_x^c = \frac{2}{1 - c\|x\|^2}, \tag{11}$$

with $\log_x^c$ its inverse. For numerical work near the boundary we recommend the Lorentz (hyperboloid) model, which is isometric and avoids the $\|x\| \to 1/\sqrt{c}$ instability.

B Notation

Symbol	Meaning
$\mathcal{M}$	product-manifold memory space
$\mathbb{H}^a_{-c_h}, \mathbb{S}^b_{c_s}, \mathbb{R}^c$	hyperbolic / spherical / Euclidean components
$d_{\mathcal{M}}$	weighted product geodesic distance
m, q	memory point, query point
η, c	reconsolidation step size, confidence
$u(m, t)$	usage potential of memory m at time t
B, b_r, U_r	total budget, region allocation, region utility